ARFAOUI BADI

Software Engineer

Core Banking & Financial Technology Systems

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Software Engineer specialized in building secure, scalable, and high-performance solutions for the banking and financial sector. Focused on core banking systems, Swift MT/MX messaging, ISO 20022 transformation, financial analytics dashboards, cybersecurity platforms, and enterprise-grade web applications — combining backend engineering, frontend architecture, security principles, and business understanding to deliver reliable solutions for critical financial environments.

SKILLS

Backend

Java · Spring Boot · Spring Cloud · REST APIs · Microservices · RabbitMQ · PostgreSQL · MySQL · MongoDB · Flask

Frontend

Angular · TypeScript · Tailwind CSS · Chart.js · Highcharts · Dashboard design

Banking & FinTech

Core Banking · Swift MT · ISO 20022 MX · MT/MX Conversion · Payment Systems · Message Validation · Financial Reporting

Cybersecurity

Secure Architecture · Vuln. Analysis · SOC / SIEM · ISO 27001 · Swift CSP · Audit & Compliance

AI & DevOps

RAG systems · Prompt engineering · Workflow automation · Docker · Nginx · Git · Linux · Keycloak

EDUCATION

Bachelor in Information Technologies
ISET Radès — Information Systems Development
2022 – 2025

Baccalaureate — Computer Science
Lycée Bouarada, Siliana
2019 – 2022

LANGUAGES

Arabic	Native
French	Fluent
English	Professional

VOLUNTEERING

Webmaster & Media Lead
IEEE ISET Radès Student Branch
May 2023 – Jun 2025

EXPERIENCE

Graduation Project — SWIFT Messaging Platform — Jan 2025 – Jun 2025
Banking Intelligence Consulting (BIC) · Tunis · Internship

- Built a microservices-based solution for SWIFT financial message analysis and management.
- Architected the platform on Spring Cloud with Netflix Eureka service discovery and Keycloak access control.

Stack: Angular, Spring Boot, Spring Cloud, Microservices, PostgreSQL, MongoDB, Flask, Keycloak, Docker

Industrial Data Analyst & Developer — Sep 2024 – Feb 2025
Société Tunisienne de Sidérurgie — El Fouladh · Tunis · Freelance

- Designed interactive Power BI dashboards for industrial analytics and KPIs.
- Built web apps to monitor and analyze machine logs for predictive maintenance.
- Integrated Python with industrial systems via WinCC and custom scripts.

Stack: Power BI, Python, Flask, Node.js, WinCC

Full-Stack Developer — Healverse — Jan 2024 – Feb 2024
Instead · Remote · Internship

- Developed a full-stack SaaS platform for the pharmaceutical industry.
- Designed a dashboard UX/UI that improved operational efficiency.

Stack: Vue.js, PrimeVue, Pinia, NestJS, Prisma, MongoDB, Docker

Web Developer Intern — Animeverse — Aug 2023 – Sep 2023
Swiftcode · Remote · Internship

- Built an e-commerce platform within a team of five engineers — first professional experience.

Stack: React, Redux, Node.js, Express, MongoDB, Bootstrap

Software Engineering Program — Mar 2023 – Apr 2023
ALX Africa · Online

- Intensive training in Linux, C, DevOps and low-level programming.

Stack: C, Python, JavaScript, Linux, Git, DevOps

FEATURED PROJECTS

Core Banking & Financial Messaging Platform
Enterprise platform to create, validate, convert, approve and analyze Swift MT and ISO 20022 MX messages with role-based access, dashboards and reporting.
Angular · Spring Boot · Swift MT · ISO 20022 · PostgreSQL

Swift Traffic Analytics Dashboard
Analytics dashboard for message traffic — categories, top clients/counterparties, currencies, FX volumes, correspondent balances and growth KPIs.
Angular · Highcharts · Chart.js · Java

MT / MX Conversion Engine
Conversion and validation engine transforming Swift MT into ISO 20022 MX via structured parsing, business mapping rules and XML generation.
Java · XML · XSD · Swift · ISO 20022

Cybersecurity & SOC Intelligence Platform
Platform to normalize security events, analyze incidents and provide audit-ready visibility for security operations.
Spring Boot · RabbitMQ · AI · SOC · OCSF